

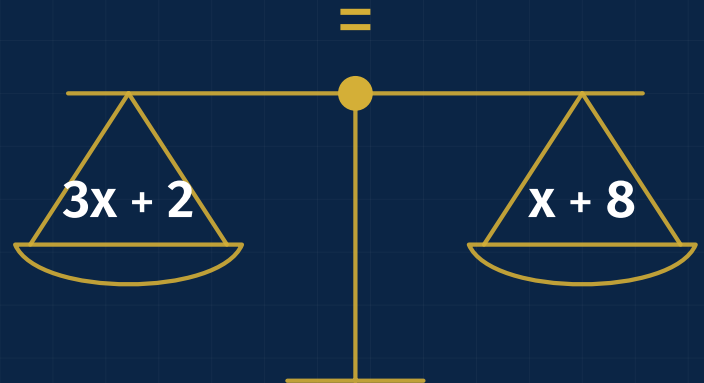


DR ESLAM AHMED
eliteigcse.com

CHAPTER 2 • ALGEBRA • TOPIC 8

Algebraic Proof

Never test numbers — use n . Translate with the dictionary, expand, then make it look like what you must prove.



6

VARIANTS

7

PAST-PAPER QUESTIONS

41

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Expression given

V2 Consecutive numbers

V3 Two different numbers

V4 Square, cube, positive

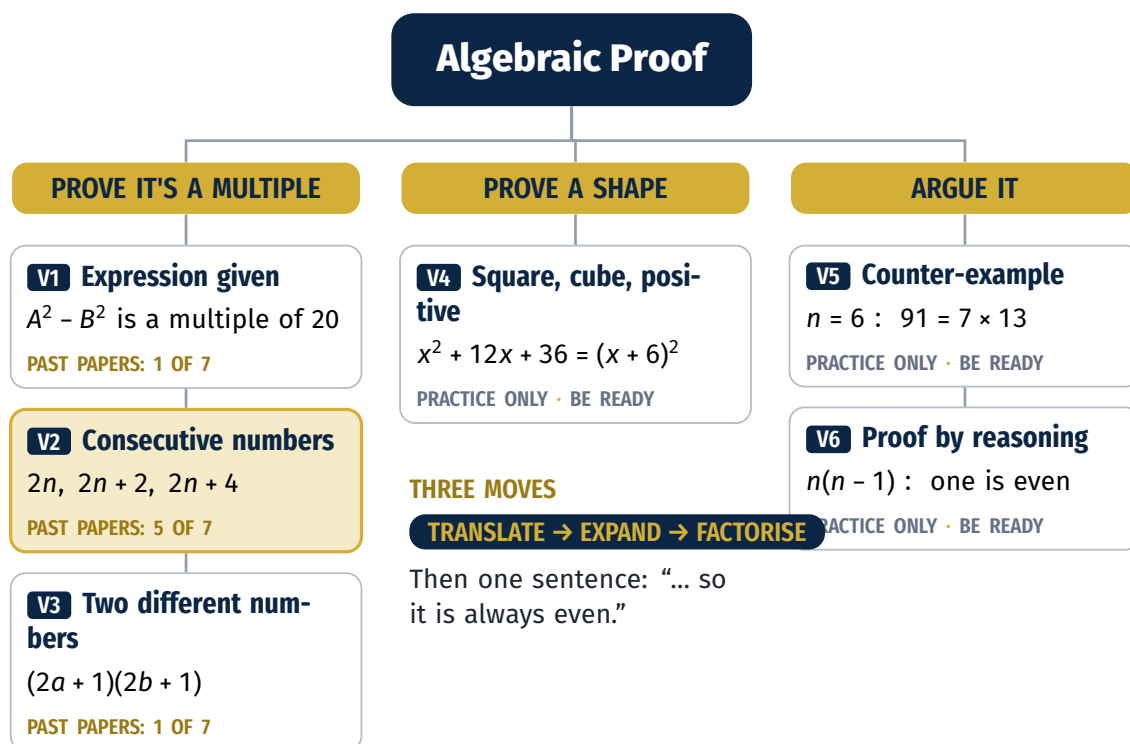
V5 Counter-example

V6 Proof by reasoning

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Algebraic Proof at a glance



Your notes


★ How often does each variant come up?

7 past-paper proof questions (Jan 2020 → Jun 2025), 3–4 marks each; 41 practice parts — grouped by type.

	PAST PAPERS (OUT OF 7)		PRACTICE PARTS
V1 Expression given	● ○ ○ ○ ○ ○ ○	1	■ 7
V2 Consecutive numbers	● ● ● ● ● ○ ○	5	■ 12
V3 Two different numbers	● ○ ○ ○ ○ ○ ○	1	■ 1
V4 Square, cube, positive	○ ○ ○ ○ ○ ○ ○	0	■ 8
V5 Counter-example	○ ○ ○ ○ ○ ○ ○	0	■ 6
V6 Proof by reasoning	○ ○ ○ ○ ○ ○ ○	0	■ 7

WHAT THIS TELLS YOU 5 of the 7 past-paper proofs are about **consecutive** numbers and their squares. Learn the dictionary — it does half the work.

BEFORE YOU START

The big idea

**A proof works for every number.**

Testing $n = 1, 2, 3$ only shows three cases. Use a **letter** that stands for **any** integer — then the result is true for all of them at once.

Dr Eslam's **dictionary** turns the words into algebra.

the words

any integer
consecutive integers
an even number
an odd number
consecutive even
consecutive odd
a multiple of 5
two **different** odd

the algebra

n
 $n, n + 1, n + 2$
 $2n$
 $2n + 1$
 $2n, 2n + 2, 2n + 4$
 $2n + 1, 2n + 3$
 $5n$
 $2a + 1, 2b + 1$

What the end must look like

even

$2(\dots)$
factor 2 out

odd

$2(\dots) + 1$
even plus one

multiple of 8

$8(\dots)$
factor 8 out

remainder 2 ($\div 8$)

$8(\dots) + 2$
multiple plus 2

a square

$(\dots)^2$
factorise to a square

a cube

$(\dots)^3$
simplify to a cube

always positive

$(\dots)^2 + \text{positive}$
complete the square

not prime

$(\dots)(\dots)$
two factors > 1

Dr Eslam's three moves

1 TRANSLATE

use the dictionary

2 EXPAND & SIMPLIFY

every bracket

3 FACTORISE TO PROVE

+ one sentence

**CALCULATOR FIRST, THEN CHECK**

Calculator first, then prove: try the statement with two or three numbers on your calculator — it shows you what the end should look like (e.g. always 8 times something). Then prove it with n . Numbers alone score 0.

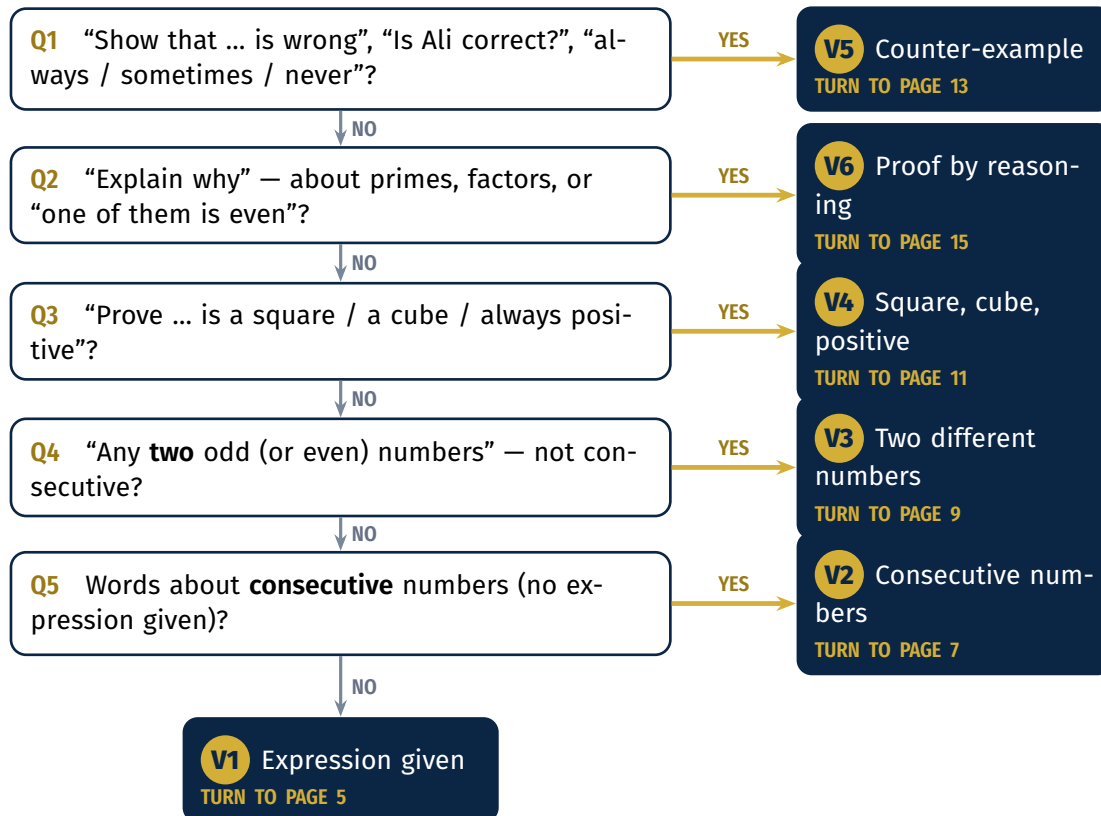
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1**VARIANT 1****Expression Given**

The expression is already in n : expand, then show the factor you need.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● **1/7**

PRACTICE QUESTIONS: 7


 Your notes

Came up in: Jan 2020 P2HR Q25

? What is it?**In plain words**

You are given an expression like $(2n + 3)^2 - (2n - 3)^2$ and must prove it is even, odd, or a multiple of something.

Skip the translating — **expand and simplify**, then **factorise out** the number you need, and say why.

IT LOOKS LIKE

$$(2n + 1)^2 - (2n + 1)$$

$$(2n + 3)^2 - (2n - 3)^2 \quad A^2 - B^2$$

$$(2n + 3)^2 - (2n - 3)^2$$

↓ EXPAND

$$4n^2 + 12n + 9 - 4n^2 + 12n - 9$$

↓ SIMPLIFY

$$24n$$

↓ FACTORISE

$$8(3n)$$

How do I do it?**1 Expand**

Every bracket — squares in full, the minus in front of a bracket hits every term.

2 Simplify

Collect: most terms cancel.

3 Factorise

Take out the number you need: $8(3n)$, $2(2n^2 + n)$.

4 Conclude

One sentence: “ $8 \times$ an integer, so a multiple of 8.”

MAKE IT LOOK LIKE THE TARGET Multiple of 8 $\rightarrow 8(\dots)$.

V1 Expression Given

VARIANT 1 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"Prove that ... is a multiple of 8 for all integers n " → **Expand, simplify, factorise 8 out**

"... is an even number" → **Show $2(\dots)$**

" N is a multiple of 5" → **Write $N = 5k$ when you need it**

WORKED EXAMPLE **PAST PAPER**

January 2020 • Paper 2HR • Q25 • 3 marks

N is a multiple of 5. $A = N + 1$, $B = N - 1$. Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20.

STEP 1 $A^2 - B^2 = (N + 1)^2 - (N - 1)^2$ ◀ substitute A and B

STEP 2 $N^2 + 2N + 1 - N^2 + 2N - 1 = 4N$ ◀ expand, simplify

STEP 3 $N = 5k : 4N = 4(5k) = 20k$ ◀ multiple of 5 → $5k$

STEP 4 $20k$ is $20 \times$ an integer, so $A^2 - B^2$ is a multiple of 20 ◀ conclude

ANSWER $A^2 - B^2 = 20k$ — a multiple of 20

✓ $N = 5: 36 - 16 = 20$

WATCH OUT

✗ $N = 5: 36 - 16 = 20$ — "proved"

✓ $A^2 - B^2 = 4N = 20k$

A number example is not a proof — it scores 0. Use letters.

EXAM TIP

The last mark is the sentence. Write it: " $20k$ is a multiple of 20."

Check yourself

calculator first • answers on page 17

1.1 Practice • Medium

Prove that $(2n + 1)^2 - (2n + 1)$ is even for all positive integers n .

Answer:

1.2 Practice • Hard

Prove that $(2n + 3)^2 - (2n - 3)^2$ is a multiple of 8.

Answer:

1.3 Practice • Hard

Prove that $\frac{6c^3 + 30c}{3c^2 + 15}$ is even (c a positive integer).

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Consecutive Numbers

A sentence about consecutive (even / odd) numbers: translate, square, simplify, factorise.

SEEN IN PAST PAPERS

●●●●●●●● 5/7

PRACTICE QUESTIONS: 12



Your notes

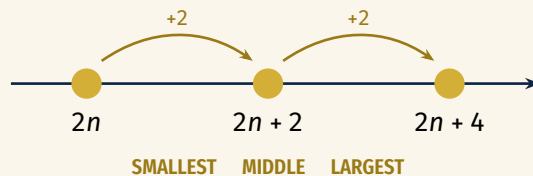
Came up in: Jan 2020 P2H Q17 · May 2021 P1H Q17 · Jan 2022 P1HR Q15 · Jan 2022 P2H Q18 · Jun 2025 P1HR Q17

? What is it?

In plain words

The question is in **words**. Use the dictionary: consecutive even numbers go up in 2s — $2n$, $2n + 2$, $2n + 4$.

Then build exactly what the sentence says (“the square of the largest minus the square of the smallest”), expand, and **factorise to match**.



IT LOOKS LIKE

consecutive even

consecutive odd

consecutive multiples of 4

How do I do it?

1 Translate

Dictionary: consecutive even $2n$, $2n + 2$, $2n + 4$; odd $2n + 1$, $2n + 3$.

2 Build the sentence

“Square of the largest minus the square of the smallest”: $(2n + 4)^2 - (2n)^2$.

3 Expand & simplify

Expand both squares fully; collect.

4 Factorise & conclude

Match the target: $8(2n + 2) = 8 \times$ the middle.

THE DICTIONARY Consecutive even/odd go up in 2s.

V2 Consecutive Numbers

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“any 3 consecutive even numbers” → $2n, 2n + 2, 2n + 4$

“two consecutive odd numbers” → $2n + 1, 2n + 3$

“... is 8 times the middle number” → **End with $8 \times (\text{middle})$**

 WORKED EXAMPLE **PAST PAPER**

May 2021 • Paper 1H • Q17 • 3 marks

Prove that, given any 3 consecutive even numbers, the difference between the square of the largest and the square of the smallest is always 8 times the middle number.

STEP 1 $2n, 2n + 2, 2n + 4$ ◀ translate

STEP 2 $(2n + 4)^2 - (2n)^2 = 4n^2 + 16n + 16 - 4n^2$ ◀ build the sentence, expand

STEP 3 $= 16n + 16$ ◀ simplify

STEP 4 $= 8(2n + 2)$ ◀ $8 \times$ the middle number ($2n + 2$)

ANSWER $16n + 16 = 8(2n + 2) = 8 \times \text{middle}$

✓ 2, 4, 6: $36 - 4 = 32 = 8 \times 4$

 WATCH OUT

✗ consecutive even: $n, n + 2, n + 4$

✓ $2n, 2n + 2, 2n + 4$

n could be odd. $2n$ makes sure every number is even.

 EXAM TIP

5 of the 7 past-paper proofs. Write the translation line first — it is the first method mark.

 Check yourself

calculator first • answers on page 17

2.1 Jan 2020 P2H

Prove that the difference between two consecutive square numbers is always odd.

Answer:

2.2 Jan 2022 P2H

Prove: the sum of the squares of two consecutive odd numbers, $\div 8$, leaves remainder 2.

Answer:

2.3 Jun 2025 P1HR

Three consecutive multiples of 4: prove $(\text{largest})^2 - (\text{smallest})^2$ is a multiple of 64.

Answer:

V3

VARIANT 3

Two Different Numbers

“Any two odd numbers” — they are not consecutive, so they need two letters.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● 1/7

PRACTICE QUESTIONS: 1



Your notes

Came up in: Jan 2021 P2HR Q15

? What is it?

In plain words

“Any two odd numbers” could be 3 and 17 — far apart. $2n + 1$ and $2n + 3$ would only cover **consecutive** ones.

So use **two different letters**: $2a + 1$ and $2b + 1$. Then expand and show the form you need.

IT LOOKS LIKE

product of two odd numbers

sum of two odd numbers



How do I do it?

1 Two letters

 $2a + 1$ and $2b + 1$ (even: $2a$ and $2b$).

2 Build & expand

 Product: $(2a + 1)(2b + 1) = 4ab + 2a + 2b + 1$.

3 Factorise

 Odd: $2(\dots) + 1$. Even: $2(\dots)$.

4 Conclude

 “ $2 \times$ an integer $+1$, so odd.”

DIFFERENT NUMBERS → DIFFERENT LETTERS

Same letter only for consecutive.

V3 Two Different Numbers

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“the product of any two odd numbers” → $(2a + 1)(2b + 1)$

“an odd number and an even number” → $2a + 1$ and $2b$

“Prove algebraically” → **Letters, expand, factorise, sentence**

**WORKED EXAMPLE** **PAST PAPER**

January 2021 • Paper 2HR • Q15 • 4 marks

Prove algebraically that the product of any two odd numbers is always an odd number.

STEP 1 $2a + 1, 2b + 1$ ◀ two different odd numbers

STEP 2 $(2a + 1)(2b + 1) = 4ab + 2a + 2b + 1$ ◀ product, expand

STEP 3 $= 2(2ab + a + b) + 1$ ◀ factorise 2 from the first three terms

STEP 4 $2 \times \text{an integer} + 1$ is odd ◀ conclude

ANSWER $2(2ab + a + b) + 1$ — always odd

✓ $3 \times 7 = 21$ odd

**WATCH OUT**

✗ $(2n + 1)(2n + 3)$

✓ $(2a + 1)(2b + 1)$

One letter only proves it for consecutive odd numbers — 1 mark at most.

**EXAM TIP**

4 marks: two letters (M1), expand (M1), the form $2(\dots) + 1$ (M1), the sentence (A1).

**Check yourself**

calculator first • answers on page 17

3.1

Dr Eslam

Prove that the sum of any two odd numbers is even.

Answer:

3.2

Dr Eslam

Prove that an even number times an odd number is even.

Answer:

3.3

Dr Eslam

Prove that an odd number minus an even number is odd.

Answer:

V4

VARIANT 4

Square, Cube, Always Positive

Simplify, then show the shape: a perfect square, a cube, or square + positive.

SEEN IN PAST PAPERS

●●●●●●●● 0/7

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 8


Your notes

? What is it?

In plain words

Square number: simplify to $x^2 + 12x + 36$ and factorise it as $(x + 6)^2$.

Cube: simplify to $8n^3 = (2n)^3$.

Always positive: **complete the square:**

$(x + \frac{1}{2})^2 + \frac{3}{4}$ — a square is never negative, plus a positive number.

IT LOOKS LIKE

$$35 + (3x + 1)^2 - 2x(4x - 3)$$

$$x^2 + x + 1 > 0$$

x	6
6x	36
x^2	6x

$$x^2 + 12x + 36 = (x + 6)^2$$

How do I do it?

1 Expand & simplify

Expand every bracket; collect to one short expression.

2 Square?

Factorise: CC Rule gives a double root $\rightarrow (x + 6)^2$.

3 Cube?

Write the number as a cube: $8n^3 = (2n)^3$.

4 Positive?

Complete the square: $(\dots)^2 + \text{positive} > 0$.

SQUARE ≥ 0 A square is never negative.

V4 Square, Cube, Always Positive

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Prove that ... is a square number" → **Simplify, factorise to $(...)^2$**

"... is a cube number" → **Simplify to $(...)^3$**

"... is always positive" → **Complete the square: $(...)^2 + \text{positive}$**

**WORKED EXAMPLE****PRACTICE**

Algebraic Proof · Medium Q8 · 4 marks

x is an integer. Prove that $35 + (3x + 1)^2 - 2x(4x - 3)$ is a square number.

- | | | |
|---------------|---|-------------------------------|
| STEP 1 | $35 + 9x^2 + 6x + 1 - 8x^2 + 6x$ | ◀ expand both brackets |
| STEP 2 | $x^2 + 12x + 36$ | ◀ simplify |
| STEP 3 | $(x + 6)^2$ | ◀ factorise: a perfect square |
| STEP 4 | $x + 6$ is an integer, so this is a square number | ◀ conclude |

ANSWER $(x + 6)^2$ — a square number✓ $x = 1$: $35 + 16 - 2 = 49 = 7^2$ **WATCH OUT**

$$\times -2x(4x - 3) = -8x^2 - 6x$$

$$\checkmark -2x(4x - 3) = -8x^2 + 6x$$

$-2x \times -3 = +6x$. Then $6x + 6x = 12x$ makes the square.

**EXAM TIP**

Always positive: $x^2 + x + 1 = (x + \frac{1}{2})^2 + \frac{3}{4}$.
Say " $(x + \frac{1}{2})^2 \geq 0$, so the whole is $\geq \frac{3}{4} > 0$ ".

**Check yourself**

calculator first · answers on page 17

4.1

Practice · Hard

Prove that $2n^2(\frac{3}{n} + n) + 6n(n^2 - 1)$ is a cube number.

Answer:

4.2

Practice · Very hard

The product of two consecutive integers is added to the larger. Prove the result is a square.

Answer:

4.3

Practice · Very hard

Prove that $x^2 + x + 1$ is always positive.

Answer:



VARIANT 5

Disprove: One Counter-Example

To show a statement is wrong, one example that fails is enough.

SEEN IN PAST PAPERS

0/7

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 6



Your notes

❓ What is it?

In plain words

To **prove** “always” you need algebra.

To **disprove** it you need just

one counter-example — a number that breaks it.

Try $n = 1, 2, 3, \dots$ until one fails, then **show why it fails**: $91 = 7 \times 13$, so not prime.

IT LOOKS LIKE

 $k = n^2 + 9n + 1$ is prime?

always / sometimes / never

n	1	2	3	4	5	6
k	11	23	37	53	71	91

PRIME, PRIME, ... $91 = 7 \times 13$

How do I do it?

1 Try values

$n = 1, 2, 3, \dots$ (and 0, negatives, fractions if allowed).

2 Find one that fails

The first value where the statement is false.

3 Show it fails

Write the working and the reason: $91 = 7 \times 13$.

ONE COUNTER-EXAMPLE IS ENOUGH ...but one example never proves “always”.

V5 Disprove: One Counter-Example

VARIANT 5 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“Show that Mo is wrong” → **One value that fails, with working**

“Is Ali correct? Give a reason” → **No + the counter-example**

“always / sometimes / never true” → **Test positive, negative and zero**

WORKED EXAMPLE **PRACTICE**

Algebraic Proof • Medium Q10 • 3 marks

$k = n^2 + 9n + 1$. Mo says “ k will be a prime number for all integer values of n from 1 to 9”. Show that Mo is wrong.

STEP 1 $n = 1, 2, 3, 4, 5 : k = 11, 23, 37, 53, 71$ ◀ all prime so far

STEP 2 $n = 6 : k = 36 + 54 + 1 = 91$ ◀ try the next value

STEP 3 $91 = 7 \times 13$ ◀ not prime

ANSWER $n = 6$ gives $91 = 7 \times 13$, not prime ✓ one counter-example

WATCH OUT

✗ “ $k = 91$, so Mo is wrong”

✓ “ $91 = 7 \times 13$, so not prime”

Show **why** it fails — the factors earn the mark.

EXAM TIP

For “always/sometimes/never”, try a negative and a fraction too: they often break statements.

Check yourself

calculator first • answers on page 17

5.1 Practice • Medium

$E = n^2 + n + 5$. Ali says E is prime for every whole number n . Is he correct?

Answer:

5.2 Practice • Medium

Show that the sum of four consecutive integers is not always divisible by 4.

Answer:

5.3 Dr Eslam

Sam says $n^2 + n + 11$ is always prime. Show Sam is wrong.

Answer:

V6

VARIANT 6

Proof by Reasoning

Factorise, then argue: consecutive numbers, primes, “one of them is even”.

SEEN IN PAST PAPERS

●●●●●●●● 0/7

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 7



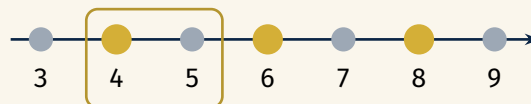
Your notes
? What is it?**In plain words**

Sometimes the proof is an **argument**, not a factor of 2. Factorise first, then explain:

■ $n(n-1)$: two **consecutive integers** — one is even, so the product is even.

■ $(2t+1)(t+2)$: two factors both > 1 — so **not prime**.

ANY TWO NEIGHBOURS:
ONE IS EVEN

**IT LOOKS LIKE**

$n^2 - n$

$2t^2 + 5t + 2$

$a^2 - b^2$ prime

How do I do it?**1 Factorise**

$$n^2 - n = n(n-1), \quad 2t^2 + 5t + 2 = (2t+1)(t+2).$$

2 Name the pattern

Consecutive integers? Two factors bigger than 1? A difference $a - b$?

3 Argue

One of two consecutive integers is even $\rightarrow$ even product.
Two factors $> 1 \rightarrow$ not prime.

4 Conclude

One clear sentence.

FACTORISE, THEN EXPLAIN

Words count here — write full sentences.

V6 Proof by Reasoning

VARIANT 6 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

$n^2 - n, n^2 + n, n^3 - n$

→ **Factorise: consecutive integers → even (or $\times 6$)**

“can never be prime”

→ **Factorise into two factors both > 1** “ $a^2 - b^2$ is prime. Why are a, b consecutive?”→ **$(a + b)(a - b)$ prime $\rightarrow a - b = 1$** **WORKED EXAMPLE** **PRACTICE**

Algebraic Proof • Very hard Q4 • 2 marks

 n is an integer greater than 1. Prove that $n^2 - n$ is never an odd number.**STEP 1** $n^2 - n = n(n - 1)$ ◀ factorise**STEP 2** $n - 1$ and n are consecutive integers ◀ name the pattern**STEP 3** one of any two consecutive integers is even, so the product is even ◀ argue**ANSWER** $n(n - 1)$ is always even — never odd✓ $n = 5$: 20 ✓**WATCH OUT**× “ $5^2 - 5 = 20$ is even, so true”✓ $n(n - 1)$: consecutive, one is evenAn example is not a reason. The argument must work for every n .**EXAM TIP**For $n^3 - n = (n - 1)n(n + 1)$: three consecutive integers — one is even and one is a multiple of 3, so a multiple of 6.**Check yourself**

calculator first • answers on page 17

6.1 Practice • Hard $a^2 - b^2 = (a + b)(a - b)$ is prime ($a > b > 0$). Explain why a and b are consecutive.

Answer:

6.2 Dr EslamProve that $n^2 + n$ is always even.

Answer:

6.3 Practice • Very hard t is a positive whole number. Explain why $2t^2 + 5t + 2$ is never prime.

Answer:

CHECK YOURSELF

Answers



Every algebra step below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1

$$4n^2 + 2n = 2(2n^2 + n),$$

even

1.2 $24n = 8(3n)$

1.3 $\frac{6c(c^2 + 5)}{3(c^2 + 5)} = 2c,$
even

2.1 $(n + 1)^2 - n^2 = 2n + 1,$
odd

2.2 $8n^2 + 16n + 10 =$
 $8(n^2 + 2n + 1) + 2$

2.3 $(4n+8)^2 - (4n)^2 = 64(n+1)$

3.1 $(2a + 1) + (2b + 1) =$
 $2(a + b + 1),$ even

3.2 $2a(2b + 1) = 2(2ab + a),$
even

3.3 $(2a + 1) - 2b = 2(a - b) + 1,$
odd

4.1 $8n^3 = (2n)^3$

4.2 $n(n + 1) + (n + 1) = (n + 1)^2$

4.3 $\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$

5.1 No: $n = 5$ gives
 $35 = 5 \times 7$

5.2 $1 + 2 + 3 + 4 = 10,$
not divisible by 4

5.3 $n = 10:$
 $121 = 11 \times 11$

6.1 $(a + b)(a - b)$ prime,
 $a + b > 1,$ so $a - b = 1$

6.2 $n(n + 1):$
consecutive, one is
even

6.3 $(2t + 1)(t + 2),$ both
factors ≥ 3

THE WHOLE TOPIC ON ONE PAGE

Algebraic Proof cheat sheet



Your notes
V1 Expression given 1/7

$$(N + 1)^2 - (N - 1)^2 = 4N = 20k$$

Expand, simplify, factorise the target out.

V2 Consecutive 5/7

$$(2n + 4)^2 - (2n)^2 = 8(2n + 2)$$

Dictionary: even $2n$, odd $2n + 1$, up in 2s.**V3 Two different 1/7**

$$(2a + 1)(2b + 1) = 2(\dots) + 1$$

Different numbers $\rightarrow$ different letters.**V4 Square, cube, positive**

$$x^2 + 12x + 36 = (x + 6)^2$$

Factorise to $(\)^2$, $(\)^3$, or complete the square.**V5 Counter-example**

$$n = 6 : 91 = 7 \times 13$$

One failure, with the reason.

V6 Reasoning

$$n(n - 1): \text{one is even}$$

Factorise, then explain in words.

**TOP 3 LOST MARKS**

1. testing numbers instead of using $n \times$
2. consecutive even as $n, n + 2 \times$
3. no final sentence $\times$

**THE TARGET SHAPES**

even $2(\dots) \cdot$ odd $2(\dots) + 1 \cdot$ multiple of 8:
 $8(\dots) \cdot$ remainder 2: $8(\dots) + 2 \cdot$ square $(\dots)^2$
 $\cdot$ positive: $(\dots)^2 + c$

BOTTOM LINE Translate (dictionary) $\rightarrow$ expand & simplify $\rightarrow$ factorise to the target $\rightarrow$ one sentence.